Quantifying relationships among populations

Advanced inference

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F-statistics

Useful to access **population similarity** and **test simple phylogenetic hypotheses**:

- Does the relationship of four test populations fits into a tree?
- Is there evidence of admixture?
- What is the admixture proportion? (limited to 2 pops and known relationships)

Do not tell

- If there has been gene-flow between various populations
- How many ancestries are required to model a population
- If unsampled lineages are required
- How a large set of populations relate to each other

Complex admixture and multi-population relationships

qpWave

qpAdm

Treemix

qpGraph

AdmixtureBayes

Complex admixture with qpWave and qpAdm

Summarize information from multiple F-statistics to make complex inferences:

- Detect the **number of “ancestries”/waves** needed to model target populations (qpWave)
- Estimate admixture **proportions** (qpAdm)

Patterson et al., 2012 (Genetics)
Haak, W. et al., 2015 (Nature)

Ancestries or waves are relative concepts. Here we use them to indicate a statistically distinguishable component given the specific population configuration tested.

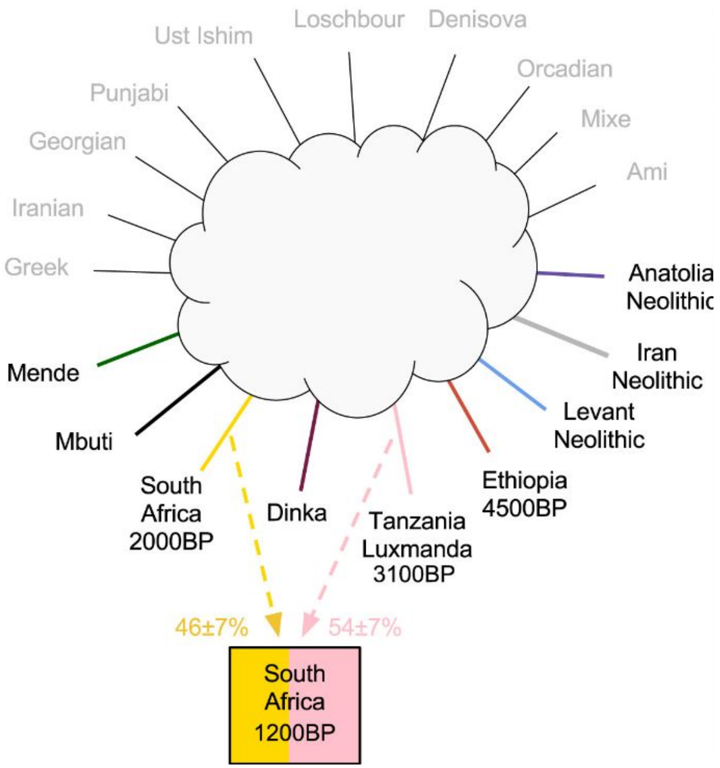
qpWave and qpAdm

Start by defining:

"**right**" populations or outgroups (not true outgroups; just differently related to the "left" populations)

"**left**" populations

- targets for qpWave
- target and references for qpAdm



Skoglund et al., 2017

qpWave and qpAdm

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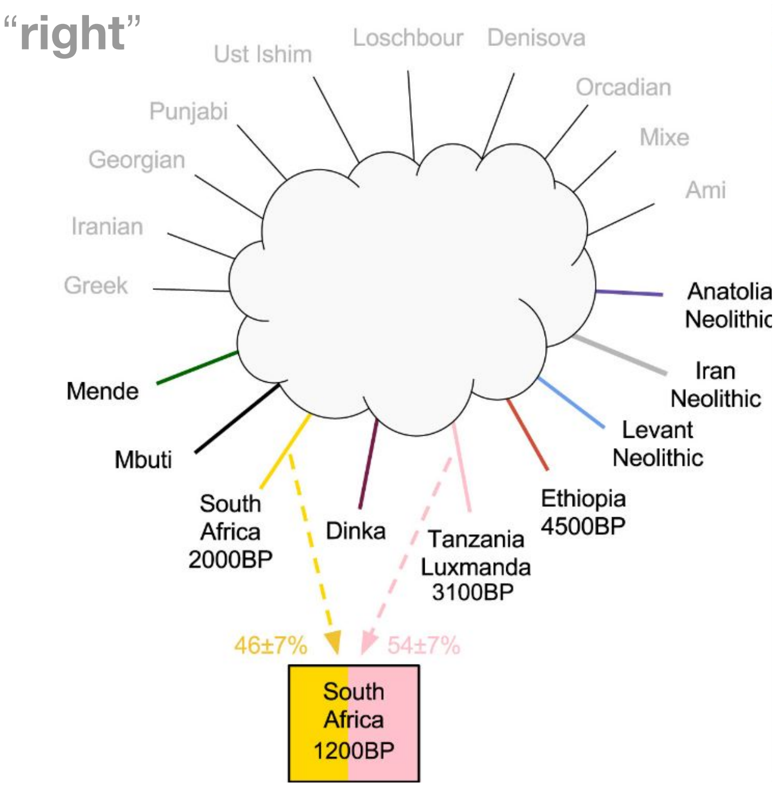

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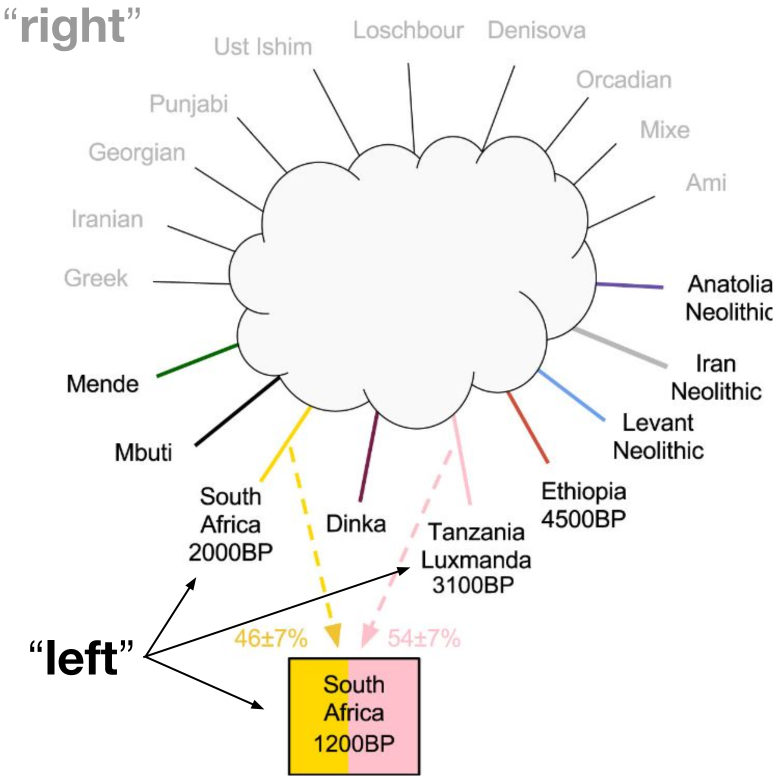


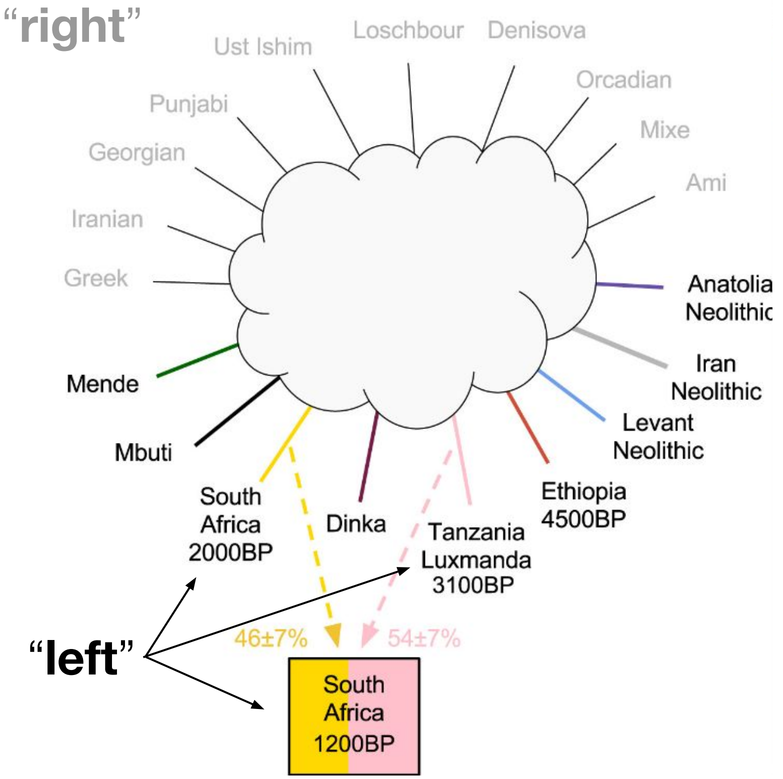
qpWave and qpAdm

Taking the first population on each side as the point of comparison, you build a $(n_L-1)\times(n_R-1)$ matrix (X) of f_4 statistics:

$$F_4(L_1,L_i;R_1,R_j)$$

If L and R form two clades, all $f_4 = 0$
If there is gene-flow, some $f_4 \neq 0$

Not all combinations need to be computed



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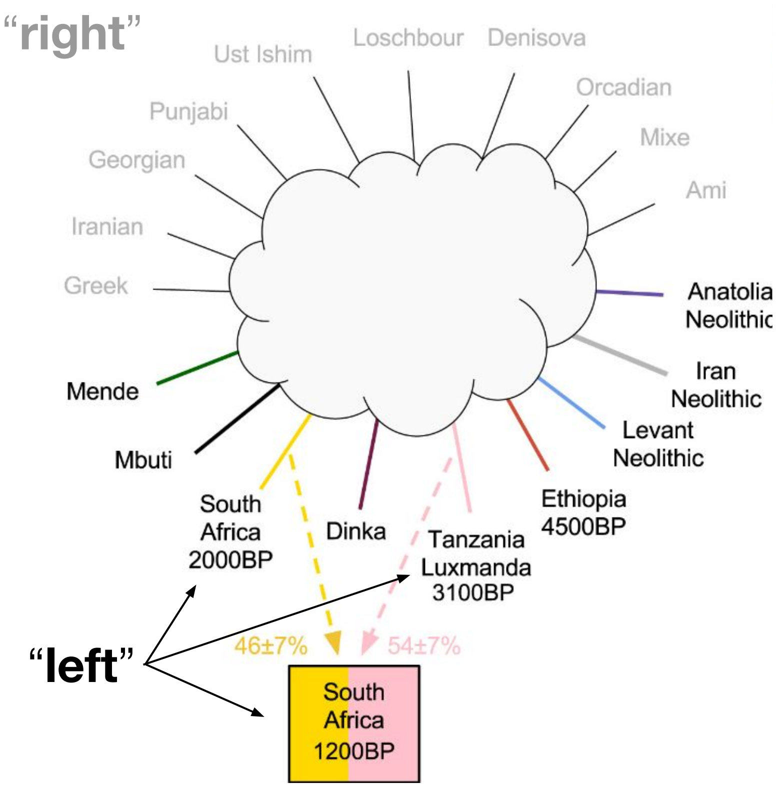
The **rank** of X tells us what is the minimum number of waves between L and R.

The highest possible rank of X is n_L-1 (assuming $n_R > n_L$)

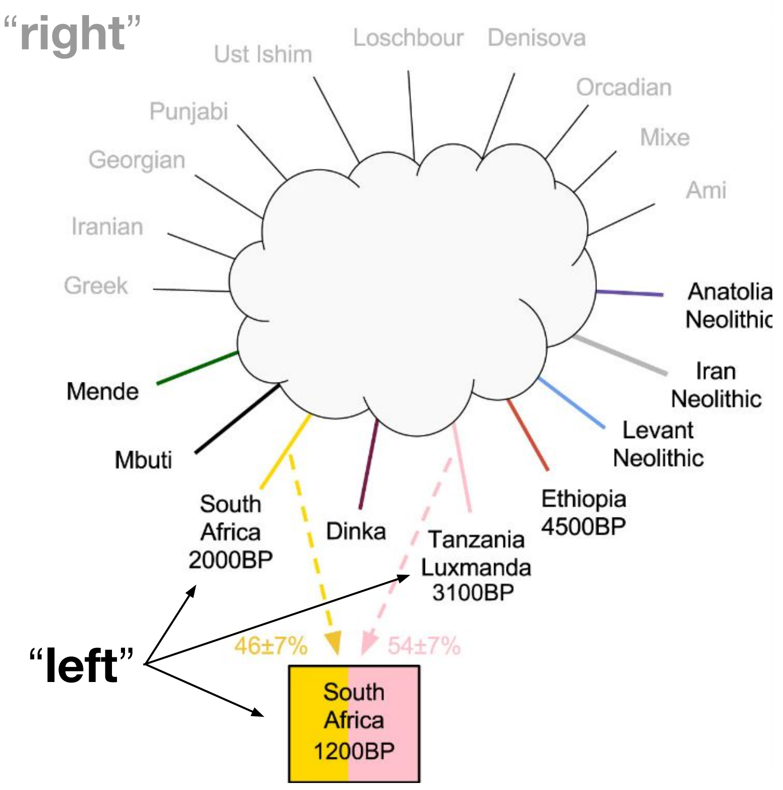


qpWave and qpAdm

Assumes that the first left population ("**target**)" is a mixture of the remaining left populations ("**references**"), and therefore not independent

Compares a model with L and R and a model with $L \cup T$ and R

A higher rank (number of waves) for the model that includes T, suggests additional gene flow between T and R

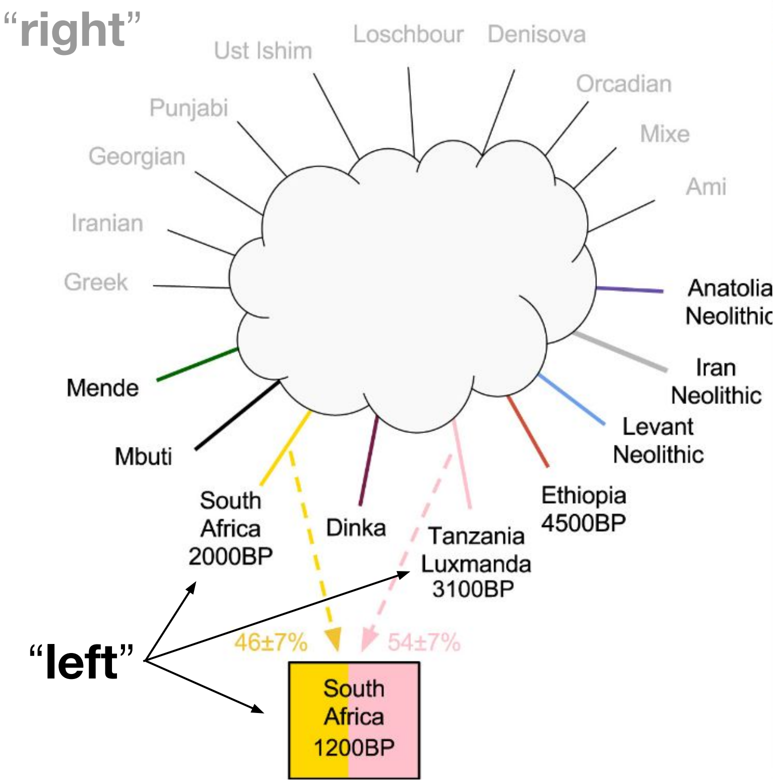


qpWave and qpAdm

If models have the **same rank**, then $\boldsymbol{T}$ **can be modeled as a combination of populations in** $\boldsymbol{L}$, and we can estimate admixture proportions

Null hypothesis: the populations in L can account for all shared drift between T and R.

A significant *p*-value rejecting this null hypothesis indicates that there has been gene flow between T and R



Summary and cautionary notes on qpWave and qpAdm

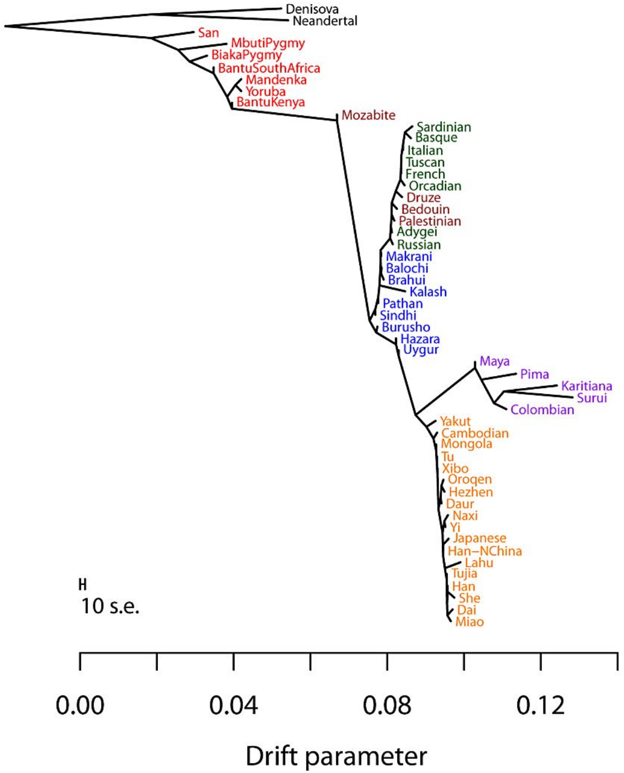
Useful for **exploring admixture** in ancient data since it can be used with:

- low coverage data
- pseudo-haploid data
- low sample sizes
- do not require imputation and phasing

Choosing left and right populations can be tricky. Inferring proportions of the true, unobserved sources depend on unverifiable assumptions: Each left population should be strictly cladal with one of the source populations

Use alternative methods (e.g., GLOBETROTTER) to **confirm your findings**

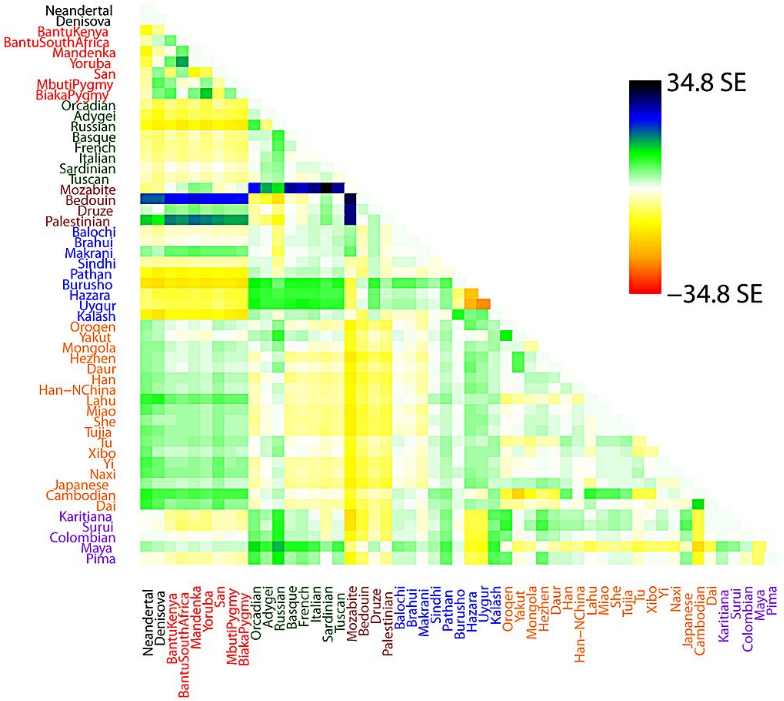
Inferring population splits and mixtures with Treemix



Given a set of allele frequencies from various populations and a specified number of admixture events, Treemix returns the **maximum likelihood tree**

Pickrell and Pritchard, 2012

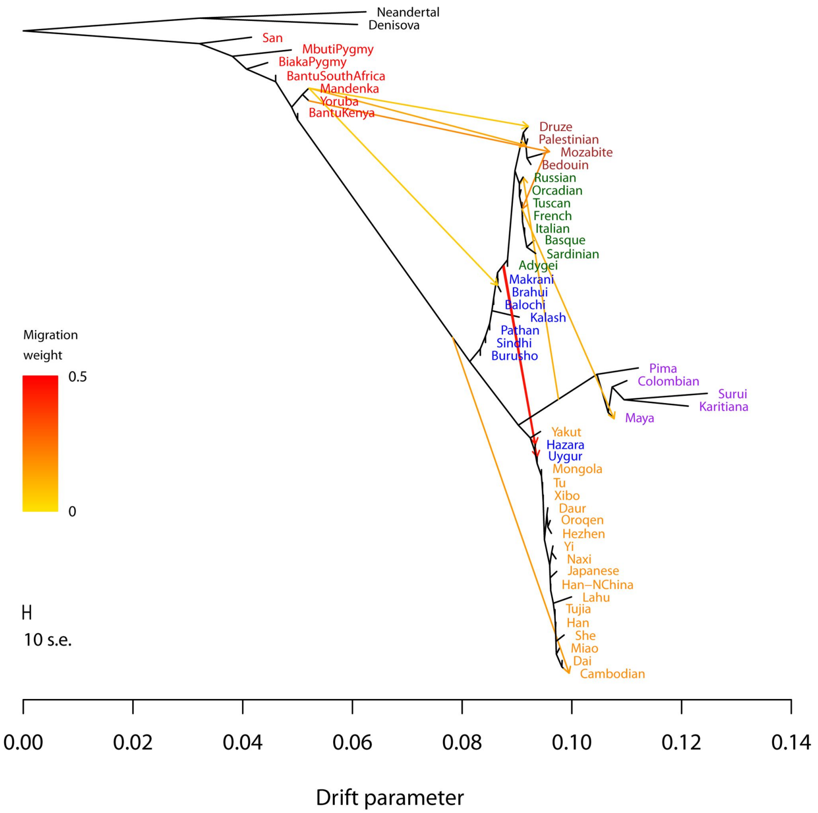
Inferring population splits and mixtures with Treemix



Model fit is evaluated through a matrix of **residuals**, which indicate pairs of populations where the model underestimates or overestimates the observed covariance

Pickrell and Pritchard, 2012

Inferring population splits and mixtures with Treemix

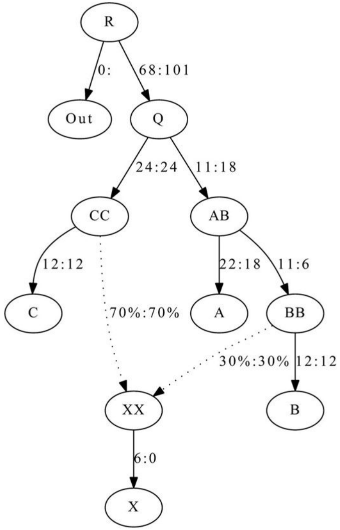


Good **starting point** to understand relationships

Direction of migration often wrongly inferred

Pickrell and Pritchard, 2012

Inferring population splits and mixtures with qpGraph



Also used to build phylogenetic trees with admixture

Compares f_2, f_3, and f_4-statistics **estimated** for the data **vs** the **expected** values given a graph

Estimates **branch lengths** (in units of genetic drift) and **mixture proportions** (weights) for a given graph

Returns poorly fitted f_4-statistics and their Z-scores, which can be useful to build alternative graphs

Patterson et al., 2012 (Genetics)

Interpretation and limitations of qpGraph

"A major use of qpGraph is to **show that a hypothesized phylogeny must be incorrect**. This generalizes our D-statistic test, which is testing a simple tree on four populations.

After fitting parameters, study of which f-statistics fit poorly can lead to insights as to how the model must be wrong.

Overfitting can be a problem, especially if we hypothesize many admixing events, but only have data for a few populations."

Patterson et al., 2012 (Genetics)

Strategy to build complex graphs

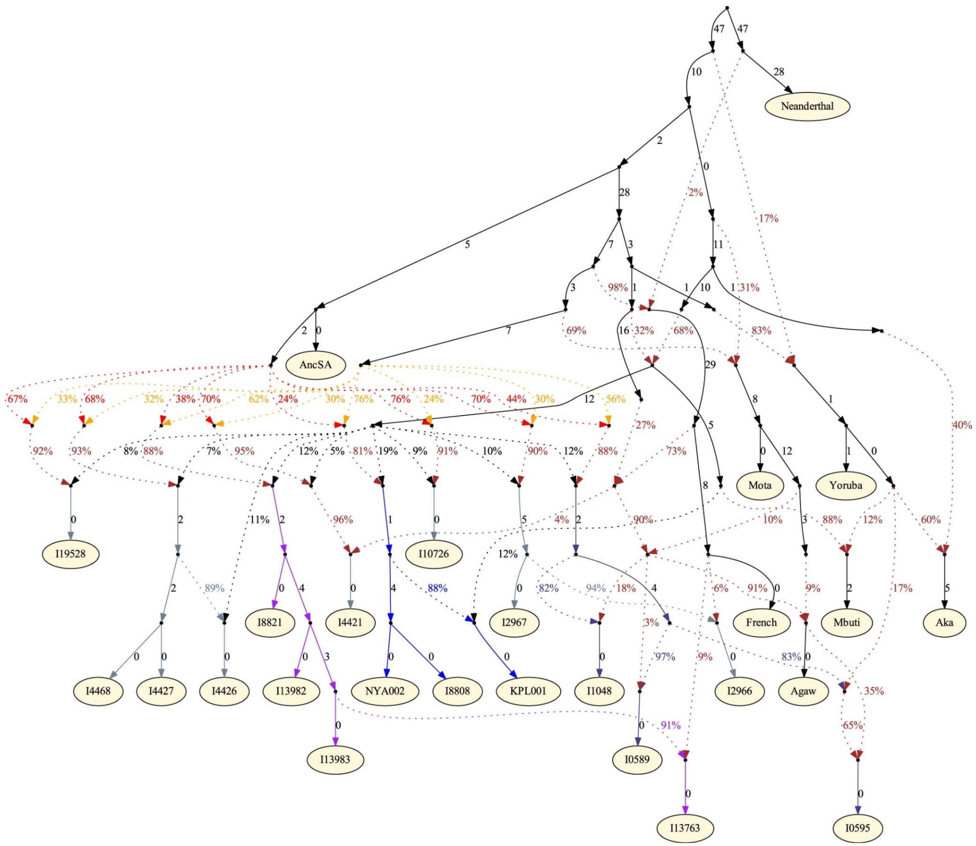
“Our usual strategy [...] is to **start with a small, well-understood subgraph and then add populations** (either unadmixed or admixed) one at a time in their best-fitting positions. This involves trying different branch points for the new population and comparing the results.”

Lipson and Reich, 2017 (MBE)

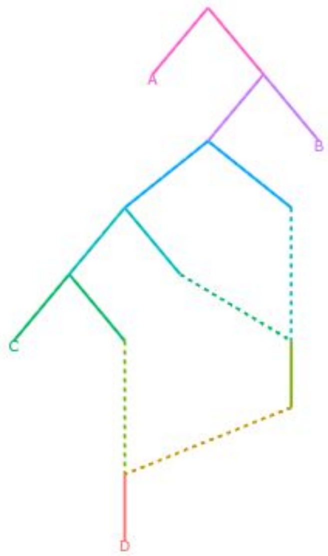
The art of building graphs

Lipson et al., 2022 (Nature)

(similarly complex examples abound in the literature)



Finding graphs with *admixtools 2*



- Automated approach to search the space of possible graph topologies
- No need for manually building the graph
- Returns the likelihood score (small score = good fit)
- Out-of-sample scores to evaluate graphs on different SNP subsets
- Bootstrap resampling of SNP blocks to compare two graphs scores are significantly different

Be aware of inconsistent results

“A model may look like it's a great fit on one particular set of SNPs, but it could look like a terrible fit on a slightly different set of SNPs. [...] This greatly diminishes our ability to meaningfully distinguish between competing models.”

https://uqrmaie1.github.io/admixtools/articles/paper.html

Be aware of inconsistent results

The "base" population (first label in input file for qpGraph) is used as outgroup in f_3 computations

If you have low quality data make sure:

- it is not highly diverged from the others
- has multiple individuals with diploid data

AdmixTools / examples.qpGraph / gr1triv

Iain Mathieson Update

Code Blame 19 lines (19 loc) · 300 Bytes

```
root     R
label    Out O
label    A   A
label    B   B
label    C   C
label    X   X
edge q   R Q
edge o   R O
edge cc Q CC
edge ab Q AB
edge c   CC C
##edge ca CC CA
edge a   AB A
edge bb AB BB
edge b   BB B
##edge ba BB BA
edge  bx BB XX
##admix XX BB CC 70 30
edge x    XX X
```

Be careful with the interpretation

"A **low score** doesn't mean that an admixture graph is a good model of reality. It just means that **we can't reject it**. [...] when there are only very **few SNPs** to work with, more models appear to fit well, but that just means that there is **less power to reject** some of these models."

Move from "What is the correct admixture graph?" to "What do all well-fitting admixture graphs have in common?"

https://uqrmaie1.github.io/admixtools/articles/graphs.html

Recommended reading

eLife

RESEARCH ARTICLE

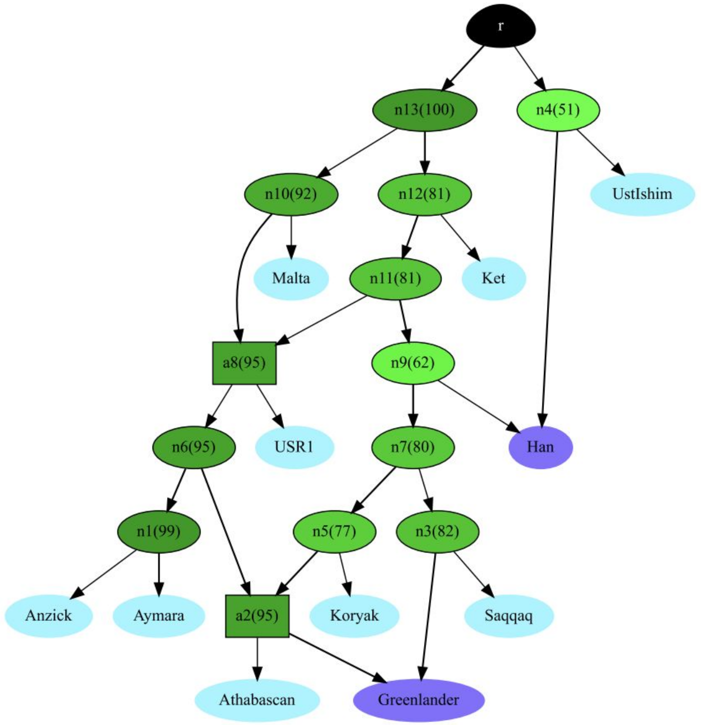
On the limits of fitting complex models of population history to *f*-statistics

Robert Maier[1*†], Pavel Flegontov[1,2*†], Olga Flegontova[2], Ulaş Işıldak[2], Piya Changmai[2], David Reich[1,3,4,5*]

[1]Department of Human Evolutionary Biology, Harvard University, Cambridge, United States; [2]Department of Biology and Ecology, Faculty of Science, University of Ostrava, Ostrava, Czech Republic; [3]Broad Institute of Harvard and MIT, Cambridge, United States; [4]Howard Hughes Medical Institute, Harvard Medical School, Boston, United States; [5]Department of Genetics, Harvard Medical School, Boston, United States

Finding graphs with AdmixtureBayes

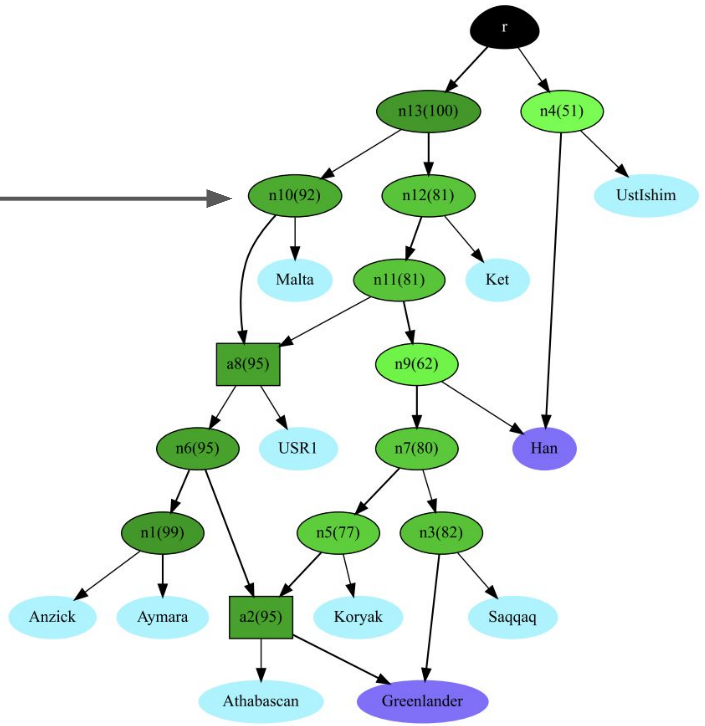
- Uses allele frequency covariances
- Fits graphs by sampling topologies from their posterior distribution (MCMC)
- **identifies highest likelihood graph**
- Explores topology uncertainty and **summarizes shared features** across high-probability graphs



Nielsen et al., 2023 PLOS Genetics

Finding graphs with AdmixtureBayes

posterior probability (92%)
that the true graph has a node
with the same descendants



Nielsen et al., 2023 PLOS Genetics

Confirm results with alternative methods

F-statistics and derived methods are based on the **same type of information** and thus **they reinforce each other**

F-statistics reduce the data to a **single estimate** of genetic distance for each population pair.

Additional information relevant for demographic history includes the joint site frequency spectrum (**SFS**) and patterns of **linkage disequilibrium**. Methods that use these can distinguish models which cannot be distinguished using f-statistics alone. They can also estimate additional parameters

Complex admixture and multi-population relationships

Admixture graphs allow:

- Exploring multi-population relationships
- rejecting hypothesis
- finding major trends

Admixtools v2 provides an automated way to find plausible graphs but:

- does not explore the full parameter space
- results are affected by ascertainment bias and data quality

Genetic data alone is usually not enough for deriving robust models → integrate different lines of evidence

Next:

Practical using *admixtools v2*